

Regression Utility of a 10,240-Byte Tabular Generator

The DOPE Project

Abstract—How much of a real-trained regressor’s improvement over a null is recovered by an auditor trained on rows from a generator under 10,240 bytes? Profile `features12_steps2048` is a compact residual: at most 12 screened inputs, width-16 tanh, a refit readout, and 2,048 AdamW steps at learning rate 0.002, with no embedding. It is not a GAN, a diffusion model, or a transformer. The population is 100 prepared PMLB regression lineages, a grouped 80/20 cut of the official training rows, seed 1729, fit seed 11, and synthetic size $4n$. The names include simulated physics and simulated dynamics. On 97 informative lineages the CatBoost median is 0.940 (95% lineage-bootstrap interval 0.914 to 0.976) against a Gaussian copula, Chow–Liu, and independent marginals. Holm-adjusted Wilcoxon tests separate these three comparisons. Against the copula the record is 86/0/11 wins/ties/losses. The linear auditor does not separate DOPE from the copula (49/0/43); the interval contains zero, and this is not an equivalence test. Unpooled blocks compare CTGAN and TVAE on 21 lineages and Forest-Flow on 6 lineages (1/0/5, 115,202,253 to 1,085,155,062 bytes). 98/100 artifacts are inside the cap, and two are over it. Official tests were not opened. TabSyn is not measured, the 100-lineage TabDDPM publication is not measured, and the full Forest-Flow publication is not measured. Fit-seed variance is not measured. Release scores are null. The distance ledger is not differential privacy or HIPAA de-identification.

Index Terms—Synthetic tabular data, retention, artifact bytes, paired tests.

I. INTRODUCTION

A generator paper often reports the metric that also selected the configuration. Those native scores live on different scales: predictive efficacy for CTGAN and TVAE [1], held-out log density for a Gaussian copula [2], a forest density for ARF [3], and a diffusion or CART objective for other families [4], [5]. This paper separates selection from the common measurement. After each method has been selected on its own objective, the shared columns are retention and the bytes charged to the artifact.

The decision those columns can support is narrow. It is whether a network small enough to ship under a 10,240-byte cap still preserves downstream regression utility on held-out validation rows. It is not a declaration that one generator is best, and it is not a certification score. Master Fitness v2 (MFS-v2), a gated geometric score, is defined in Section III. It is null on every method here, because the six components and the hard-gate map are not all present.

The contributions are:

- 1) The displayed residual and the generative procedure that samples a row from that artifact (Section III, Fig. 1, (1)). That procedure is not the production default and it does not use the 4,168-dimensional router.

- 2) Retention and charged bytes on the validation panel, with lineage-bootstrap intervals, and the algebra of a retention value (Fig. 2).
- 3) Paired tests inside unpooled blocks: the density block, the neural block, and the 6-lineage Forest-Flow confirmation, including the linear non-separation and the Forest-Flow lead (Fig. 3).

a) Scope.: TabSyn is not measured. The 100-lineage TabDDPM publication is not measured. The full Forest-Flow publication is not measured. The pre-specified privacy-attack panel is not measured. Fit-seed variance is not measured. The full ablation grid is not measured. The fits use one fit seed, 11, and coresets of 72 to 800 rows. Official tests stay sealed. MFS-v2 is null. PTF-v1, the release KPI whose hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout, is null. The repository does not expand those three letters. A release at the 10,240-byte tier is a separate claim from the byte cap, and it is not made here. BeyondArena, a separate family-fit check outside this PMLB panel, is not pooled with the density, neural, or Forest-Flow blocks. An unexecuted method is not a win for DOPE. Later sections point here instead of repeating this list.

II. RELATED WORK

Tabular synthesis has moved from parametric copulas [2] and Chow–Liu trees [6] through conditional GANs, including TableGAN, and variational autoencoders [1], [7] to diffusion and score-based models [4], [8]–[14]. Autoregressive and language-model generators [15], [16], prior-data fitted generators [17], [18], CTAB-GAN and CTAB-GAN+ [19], [20], and graphical neural generators [21] target the same mixed-type tables. ARF estimates density with a forest [3]. CART synthesis is the synthpop line [5]. A recent survey collects these design choices [22]. This paper runs the Gaussian copula, a Chow–Liu baseline written for this comparison, independent marginals written for this comparison, CTGAN, TVAE, and a 6-lineage Forest-Flow confirmation. TabSyn, TabDDPM on the 100-lineage panel, STaSy, CoDi, TabDiff, CTAB-GAN+, GReaT, REaLTabFormer, and TabPFGen are cited and not measured. SMOTE [23] was not run. This panel scores regression retention, not classification oversampling.

Differentially private Bayesian networks and marginal synthesizers [24]–[27] optimize a privacy budget. Membership inference and shadow-model attacks are a separate detection literature [28]–[30]. Formal differential privacy is a different claim from an empirical resemblance score [31], [32]. Utility reviews

separate marginal fidelity from downstream prediction [33], [34]. Alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports [35]–[37]. The fidelity results add alpha-precision, beta-recall, a distance-to-closest-record share, and a DOMIAS AUC. ARF is not in that fidelity table. The displayed-profile publication median is 0.840. The pre-specified privacy-attack panel remains not measured. Nothing in those fields is a formal privacy guarantee or a HIPAA de-identification claim.

Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors [38]–[40]. TabZilla and TabArena show how brittle a single split can be [41], [42]. The regression lineages come from PMLB [43], [44]. Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative [45], [46]. This paper reports the frequentist procedure: a Friedman test and a Nemenyi critical difference on the common set, plus paired Wilcoxon tests. The Bayesian signed-rank procedure is not computed.

III. METHOD

A. Displayed generator

The displayed profile is `features12_steps2048`. It is a compact neural residual, not a GAN, a diffusion model, or a transformer. Fig. 1 is the pipeline. Training screens at most 12 inputs. A hidden layer of width 16 maps them through $\tanh$. AdamW, learning rate 0.002, runs for 2,048 steps on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen. The intercept, the linear skip, and the hidden readout are refit on those frozen features. Coefficients whose absolute value is below 10^{-5} are dropped from the skip. For a selected input vector x_S ,

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

There is no embedding table in (1). A separate dataset-action router uses a vector of dimension $856 + 24 \times (128 + 10) = 4,168$. That vector does not generate the rows in this paper.

Equation (1) is the conditional mean, not the joint draw. Sampling for this profile uses the symbolic kernel stored in the artifact. It does not use the separate neural-joint sampler. The dependence stored with an artifact is one choice from the compile search: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, or a two-component mixture. This paper does not treat the displayed artifacts as one dependence family.

Training, not sampling, screens inputs by absolute rank correlation with the target and keeps at most 12. The artifact stores a marginal and a transform for each feature, the selected dependence, the residual, a noise model, and a per-feature missingness rate. A draw samples a latent vector from the stored dependence, maps each coordinate through the standard normal distribution function, inverts the marginal, and inverts the feature transform. The residual then supplies $\hat{y}$. When the training target is not entirely zeros and ones, the stored noise is heteroscedastic: a linear fit of the log absolute residual on up to four screened features, with the scale clamped between the lower and upper deciles of those absolute residuals. The

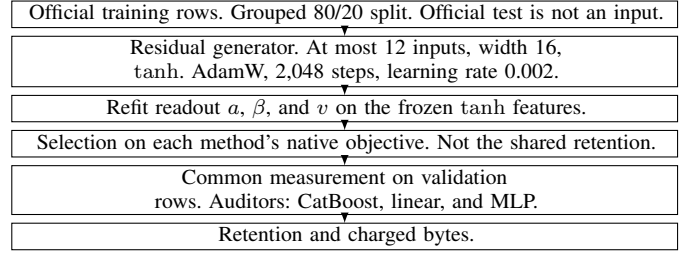


Fig. 1. Displayed pipeline. The official test is not an input.

sampled target adds Gaussian noise of that scale and then clamps the sum to the unit interval of the training transform. It is not returned in the original units of the source column. Each feature is then replaced, independently, by a missing mark with its stored probability. The charged byte count is the length of the encoded artifact. Before that encoding, the compiler quantizes the marginals, the dependence, the residual, and the noise model.

The profile was fixed before the shared retention was inspected. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every candidate record withholds a release certificate. The compile record is not a release certificate. No global DOPE family has been selected. The production default, used when this research profile is not selected, clamps the hidden width from 4 to 12 and trains for 96 steps. It is not this result. Other shapes exist and are not this measurement: micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The scored campaign did not store a per-step loss. A later logged replay did. The supplement plots that replay. The replay files are not substituted for the scored artifacts, and a lower hidden-basis loss at 8,192 steps does not replace the displayed profile. The full width-and-step grid is not measured.

B. Retention and MFS-v2

Let L_{null} , L_{real} , and $L_{\text{synthetic}}$ be the auditor's loss on held-out validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Those held-out rows come from the official training file. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

Rearrangement, with r the retention, is algebra on those three losses:

$$L_{\text{synthetic}} = L_{\text{real}} + (1 - r)(L_{\text{null}} - L_{\text{real}}).$$

Invented losses 1, 0.01, and retention 0.94 give a synthetic-trained loss of 0.0694, which is 6.94 times the real-trained mean squared error. The CatBoost median in this paper is not this example. A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat that auditor. A negative value means it did worse than

the null. A lineage is informative when the real improvement is at least 0.01 times the absolute null loss and is not zero. Otherwise the lineage is omitted. It is not entered as zero. The three auditors are CatBoost [39], ordinary linear regression, and an MLP. The lineage score is the median of sample seeds 101, 211, and 307. The panel score is the median of those lineage scores. Intervals below are percentile lineage-bootstrap intervals, 10,000 resamples of lineages, with the three seeds kept inside the lineage median. They are not per-seed whiskers.

MFS-v2 is a different object. With component $c_i \in [0, 1]$, weight $w_i > 0$, $\epsilon = 10^{-6}$, and $W = \sum_i w_i$,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i)\right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside $[0, 1]$. Otherwise the scalar is null. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05. A privacy failure removes the score through a gate. Utility transfer inside (3) is a clamped one-sided lower bound, not the median of (2). Compactness is $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$. Crossing 10,240 bytes fails the byte gate, so the scalar stays null. An artifact over the cap is a failed gate, not a retention of zero. No artifact in this paper has a measured MFS-v2 value. Copying a retention number into that scalar would publish a value the contract refuses to emit. The supplement applies (3) to invented components, including a zero compactness term, and lists the null column. Those rows are arithmetic, not measurements.

IV. EXPERIMENTAL SETUP

A. Partitions and provenance

The panel is the rights-cleared PMLB regression collection [43], [44]. Of these prepared lineages, 20 names begin with feynman, which the PMLB catalog describes as synthetic physics, 14 begin with strogatz, which that catalog describes as simulated dynamics, 25 contain the Friedman-style fri marker, 6 contain bng and are Bayesian-network generators, and 35 are the remainder. This paper does not re-verify which of the remainder are observational measurements. Distinct lineage identifiers are not a claim of independent source families. Related names, including the cpu_small lineages, stay inside the one-row-per-lineage bootstrap. The resampling unit is the lineage, so related variants move together only by chance. A source-family interval is not published, because a verified parent map is not in the ledger. The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (`check_nonoverlap` raises on that overlap). Another 3,755 catalog entries lack a redistribution grant and are not backfilled. The prepared cut is a grouped 80/20 of the official training rows, seed 1729. Worker directories hold `train.csv` and `validation.csv`. They do not hold `test.csv`. Official tests stay sealed, and final evaluation is not authorized. The fit view has 72 to 800 rows. Sources above the quota use a hybrid coresets that keeps the observed feature and target range. The

supplement records that draw. Held-out validation rows are the grouped holdout of the official training file, and those source rows include the simulated generators named above. Table I is the lock and the file-name count. Test-file contents were not read.

Fit seed 11 is the population actually measured. Sizes n and $4n$ are both stored. The tables below are $4n$. The Gaussian copula row is Copulas 0.14.1 `GaussianMultivariate`, selected by held-out mean log density. CTGAN and TVAE are the author implementation, selected by the mean of linear and MLP training-mean skill on validation rows, the same skill defined in Section V. Chow-Liu uses eight bins and Laplace smoothing 1.0. It is an implementation written for this comparison, as are the independent marginals. The ARF watch log's last count is 799 closed and 799 ok out of 800 planned, then a sealing record. That log is not a retention panel. The displayed-profile ARF publication median is 0.840. Forest-Flow on 6 lineages uses the author regression objective on its frozen grid. Native objectives are not ranked across methods.

B. Statistics

Comparisons are paired on the lineages where both methods have a complete informative group. The tests are two-sided Wilcoxon signed-rank tests on the nonzero paired differences, and one-sided sign tests for a positive difference, with a matched-pairs rank-biserial correlation. Holm correction is applied inside a family, separately for each test, and is not pooled across families. On CatBoost the Holm sign-test values are 2.15×10^{-15} , 6.72×10^{-24} , and 4.95×10^{-27} . The density family has nine tests (three comparators times three auditors). The neural family is CTGAN and TVAE times the auditors. The Forest-Flow family is separate and has six lineages. A Friedman test and a Nemenyi critical difference use the four density methods on their common complete-informative set, with rank 1 assigned to the higher retention. The critical-difference constant for four methods at $\alpha = 0.05$ is 2.569. The interval resamples the informative lineages of this panel. Each lineage contributes the median of sample seeds 101, 211, and 307 from the single fit at seed 11. The interval does not include a new fit, a new split, a new population, or a change of auditor. These are descriptive validation tests. They are not the release KPI in Section I-0a, and they do not authorize a release at the 10,240-byte tier. MFS-v2 and PTF-v1 stay null.

C. Compute

Table II is receipt wall-clock and charged bytes. Fit and sample columns are `elapsed_seconds` when the receipt stores that timer. A timer, device, or byte field the receipt does not store is not recorded. The hardware cell names the GPU models recorded on the receipts. It does not name a machine. CPU model is not recorded on every row. The TabSyn cost row is not measured. The TabDDPM cost row is not measured. The supplement states the replay walls, the Forest-Flow sums, and the unsplit density attempt timers.

V. RESULTS

Table III keeps, for each block and auditor, the tuned baseline with the highest paired median retention. The difference is

TABLE I
TEST-SPLIT PROVENANCE. FILE-NAME COUNTS ONLY.

Item	Record
Catalog SHA-256	ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120
Eligible / prepared / overlap exclusions	104 / 100 / 4
Unknown-rights catalog entries held out	3755
Split	official_test_grouped_training_80_20, seed 1729
Overlap lineage ids	590210721973399a, 9fe58a656fdb458a, d06a584f07d0acd5, d0ba629ed07b5d63
Final evaluation authorized	false
S3 worker directories / train.csv / validation.csv	100 / 100 / 100
test.csv under S3 workers	0
BeyondArena workers / worker test.csv	5 / 0
Evaluator test.csv files, contents not read	5

TABLE II
COMPUTE AND ARTIFACT COST. MISSING RECEIPT FIELDS ARE NOT RECORDED.

Method	Fit s	Sample s	RAM / VRAM	Hardware	CPU h, GPU h	Bytes
DOPE	7.25(13.5)	not recorded	920.7/ 1067	one NVIDIA GeForce RTX 4090 and two NVIDIA GeForce RTX 3090 GPUs	not recorded, 0.193	1,895.5(758–22,233)
Gaussian copula	not recorded	not recorded	not recorded	host not recorded	not recorded	5,464.5(892–833,439)
Chow–Liu	not recorded	not recorded	not recorded	host not recorded	not recorded	29,865(3,348–2,035,888)
Independent marginals	not recorded	not recorded	not recorded	host not recorded	not recorded	2,549.5(862–36,672)
CTGAN	213.5(268.1)	186.7(197.0)	1557.8/ 1101	NVIDIA GeForce RTX 4090	not recorded, 2.52	741,834(528,075–1,428,695)
TVAE	214.2(253.2)	186.1(196.6)	1252.9/ 1069	NVIDIA GeForce RTX 4090	not recorded, 2.45	317,141.5(157,578–741,659)
Forest-Flow	185.9(380.6)	342.4(511.7)	6112.0/ 3181	NVIDIA GeForce RTX 4090	not recorded, 2.44	325,848,913.5(115,202,253–1,085,155,062)
ARF	not recorded	not recorded	not recorded	GPU model not recorded	not recorded	1,627,718.5(21,658–34,331,533)
TabSyn	not measured	not measured	not measured	not measured	not measured	not measured
TabDDPM	not measured	not measured	not measured	not measured	not measured	not measured

DOPE minus that baseline. Its interval is the paired lineage bootstrap already used in the block tables. W/T/L counts lineages with a positive, zero, or negative difference. Holm p is the adjustment inside that block’s existing family. It is not a new family, and the blocks are not pooled. A positive median with a small p -value still includes lineage losses. The paired median difference is not the difference of the two medians. 0.224 is not 0.940 minus 0.656.

A. Density block

Fig. 2 plots retention against $\log_{10}$ charged bytes for the three auditors. Circles are DOPE, squares the Gaussian copula, triangles Chow–Liu, and diamonds the independent marginals. Filled marks are lineages. Open marks are that method’s median and its lineage-bootstrap interval on the lineages it completed. The table intervals are the paired subsets. For CatBoost those two DOPE sets coincide. Paired differences outside $[-1.5, 1.5]$ are counted in the panel title and omitted from that frame. The retention axis stops at 1.6. The dashed line is the cap. Neural and Forest-Flow artifacts are not on this figure.

Table IV is the paired density block. Under CatBoost, DOPE’s median paired differences are positive against the Gaussian copula (0.224, interval 0.149 to 0.290), the Chow–Liu tree, and the independent marginals. The Holm-adjusted p -values are 1.35×10^{-13} , 4.85×10^{-16} , and 1.17×10^{-16} . The rank-biserial correlations are 0.890, 0.978, and 0.999. The Gaussian copula is the strongest density baseline. Wins, ties, and losses against it are 86/0/11 under CatBoost and 65/0/10 under the MLP. On the 97 lineages common to all four methods, the Friedman p -value is 2.71×10^{-44} and the critical difference

is 0.476. All six CatBoost pairs exceed that difference. Fig. 3 shows the sorted paired differences with the same four marks. The horizontal axis is the retention difference. The bar is the critical difference.

The linear auditor does not separate DOPE from the Gaussian copula (49/0/43; Holm 0.373; difference 0.00366, interval -0.00438 to 0.012 ; rank-biserial 0.107). The interval contains zero. That non-separation is not an equivalence test. Ranks are 1.533 and 1.598 against critical difference 0.489. Chow–Liu and the independent marginals remain separated under the linear auditor and the MLP. A CatBoost separation is not a separation under every auditor.

Table V is the median raw loss on the same informative lineages. RMSE is the square root of the stored mean squared error. Training-mean skill is one minus that loss divided by the training-mean null loss. Conventional held-out R^2 uses the mean of the evaluation targets and was not stored for these rows. Retention is the median of the per-lineage ratios, not the ratio of these medians. CatBoost reads RMSE 0.053 and 0.081, with training-mean skill 0.881 and 0.792.

Table VI counts artifacts at or under the cap. The two displayed DOPE artifacts over the cap are 4544_Geographical OriginalofMusic at 22,233 bytes and 588_fri_c4_1000_100 at 19,019 bytes. 678_visualizing_environmental has a fit whose CatBoost group is not informative, so the CatBoost median uses 97 lineages while the byte count uses 100. Independent marginals pass the cap more often than the Chow–Liu baseline and still sit near retention zero. Smallness alone is not the result. The displayed byte range runs from 758 to 22,233.

On the density cells, the displayed profile has median classifier two-sample AUC 0.517 (interval 0.504 to 0.532,

TABLE III
STRONGEST TUNED BASELINE AT $4n$. W/T/L COUNTS LINEAGES. HOLM STAYS INSIDE THE BLOCK.

Block	Auditor	Baseline	n	Difference	W/T/L	Holm p
Density	CatBoost	Gaussian copula	97	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
Density	Linear	Gaussian copula	92	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Density	MLP	Gaussian copula	75	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
Neural	CatBoost	TVAE	21	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Neural	Linear	TVAE	21	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
Neural	MLP	TVAE	19	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}
Forest	CatBoost	Forest-Flow	6	-0.027 [-0.308, 0.00804]	1/0/5	0.375
Forest	Linear	Forest-Flow	5	0.012 [-0.260, 0.302]	3/0/2	0.812
Forest	MLP	Forest-Flow	4	-0.032 [-0.227, -0.010]	0/0/4	0.375

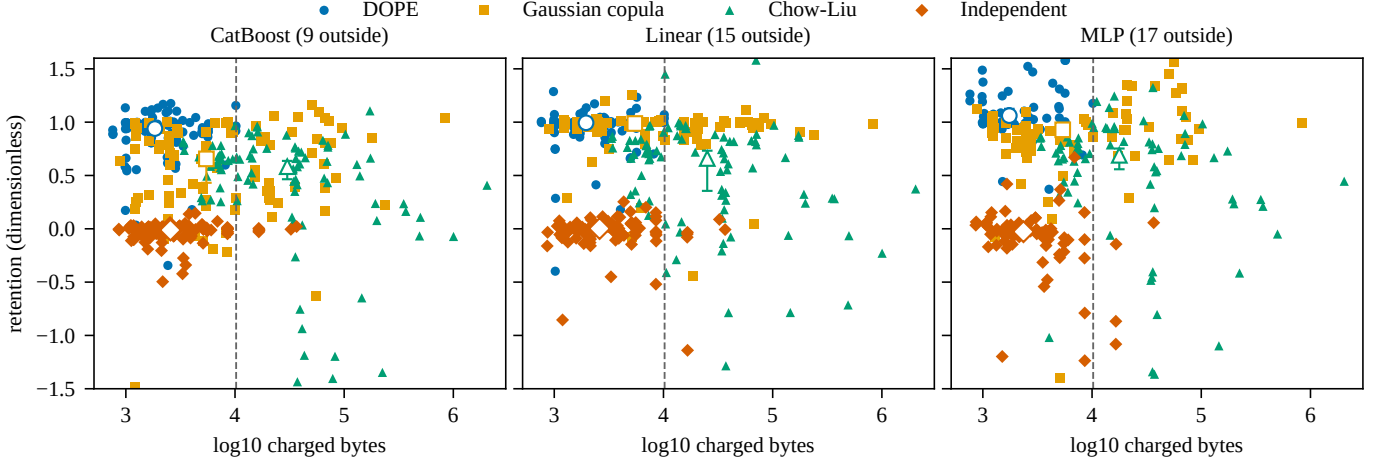


Fig. 2. Density-block retention at $4n$ against charged bytes. Three auditors. The dashed line is the 10,240-byte cap.

TABLE IV
DENSITY BLOCK AT $4n$. W/T/L COUNTS LINEAGES.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	Gaussian copula	97	0.940 [0.914, 0.976]	0.656 [0.431, 0.803]	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
CatBoost	Chow-Liu	97	0.940 [0.914, 0.976]	0.586 [0.465, 0.637]	0.337 [0.282, 0.451]	94/0/3	4.85×10^{-16}
CatBoost	Indep. marginals	97	0.940 [0.914, 0.976]	-0.012 [-0.020, -0.00649]	0.956 [0.918, 0.980]	96/0/1	1.17×10^{-16}
Linear	Gaussian copula	92	0.994 [0.981, 0.996]	0.989 [0.968, 0.994]	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Linear	Chow-Liu	92	0.994 [0.981, 0.996]	0.661 [0.356, 0.738]	0.340 [0.244, 0.425]	86/0/6	7.12×10^{-15}
Linear	Indep. marginals	92	0.994 [0.981, 0.996]	0.00206 [-0.023, 0.00583]	0.988 [0.975, 1.001]	92/0/0	5.70×10^{-16}
MLP	Gaussian copula	75	1.061 [1.024, 1.125]	0.928 [0.865, 0.953]	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
MLP	Chow-Liu	75	1.061 [1.024, 1.125]	0.692 [0.582, 0.754]	0.530 [0.350, 0.651]	74/0/1	2.11×10^{-13}
MLP	Indep. marginals	75	1.061 [1.024, 1.125]	-0.028 [-0.047, -0.013]	1.104 [1.047, 1.179]	75/0/0	2.11×10^{-13}

TABLE V
RAW AUDITOR MEDIANS ON THE INFORMATIVE LINEAGES.

Auditor	n	RMSE real	RMSE synth.	training-mean skill, real	training-mean skill, synth.
CatBoost	97	0.053	0.081	0.881	0.792
Linear	92	0.105	0.112	0.585	0.551
MLP	75	0.098	0.083	0.666	0.785

$n = 98$), mean marginal Kolmogorov-Smirnov distance 0.114 (0.103 to 0.135), and pairwise correlation fidelity 0.953 (0.945 to 0.961). A two-sample AUC near one half means a linear-time classifier has trouble telling the synthetic rows from the held-out validation rows. Table VII adds alpha-precision, beta-recall, the unbalanced DCR share, and a DOMIAS AUC. The lower block is the neural lineages only. Holm families are 12

TABLE VI
CHARGED BYTES AT $4n$. FOREST-FLOW IS NOT POOLED.

Method	Median bytes	Within cap
DOPE reference	1,895.5	98/100
Gaussian copula	5,464.5	62/100
Chow-Liu	29,865	31/100
Independent marginals	2,549.5	94/100
CTGAN	741,834	0/22
TVAE	317,141.5	0/22
Forest-Flow	325,848,913.5	0/6

and 8, separate from retention. The DCR column is the fraction of synthetic queries whose nearest reference is in the fit view, with ties scored one half. A balanced variant was computed in the harness and is not this column. Under an independent draw, the unbalanced share sits near the fit-view fraction of the

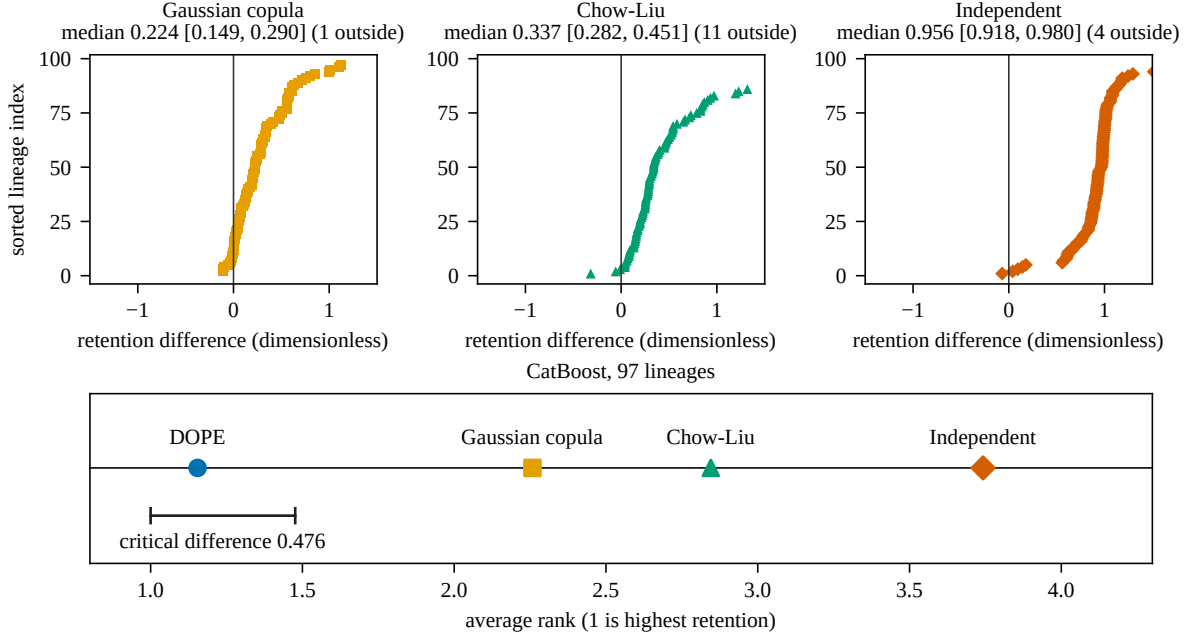


Fig. 3. CatBoost paired differences, sorted, and the Nemenyi scale. Rank 1 is the higher retention. The bar length is the critical difference.

pooled references. The grouped split keeps about four fifths of those rows in the fit view, so the baseline is near four fifths, not one half. Higher values of that share, and higher DOMIAS AUC, sit nearer the fit view and are not a quality win. DOMIAS here is the isolated kernel density-ratio equation from the pinned author expression, evaluated on this numeric projection. It is not the original DOMIAS training pipeline and it is not a membership-inference certificate. The DOMIAS AUC stays near one half. The density DCR contrast against independent marginals clears Holm in that worse direction. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. This is not differential privacy or HIPAA de-identification.

B. Neural block

Table VIII is not pooled with Table IV. On the 21 CatBoost lineages, DOPE’s median is 0.953, CTGAN’s is -0.081 , and TVAE’s is 0.669. TVAE is the stronger neural baseline. Wins, ties, and losses against TVAE are 19/0/2, 20/0/1, and 17/0/2. The median favors DOPE, and DOPE still loses lineages. Every successful CTGAN and TVAE artifact in this byte count is over the cap (0/22 and 0/22). Runs that stopped at the recorded time limit are omitted. They are not counted as failures of a finished fit.

C. Forest-Flow

Table IX is a 6-lineage confirmation, not pooled with the density or neural blocks. The Forest-Flow CatBoost median is 1.003 (interval 0.926 to 1.026) and the paired DOPE median is 0.940. Wins, ties, and losses are 1/0/5 under CatBoost, 3/0/2 under the linear auditor, and 0/0/4 under the MLP. DOPE is lower on every paired MLP lineage. The Holm tests are not significant at this count, so this is not a certified win for either

method. Charged Forest-Flow artifacts run from 115,202,253 to 1,085,155,062 bytes (median 325,848,913.5) against a DOPE median of 2,067 on the same lineages. MFS-v2 on this block is null.

Table X is a different ledger: the published matched-population receipt for ARF, TabDDPM, TabSyn, and Forest-Flow. Rows are the displayed profile at $4n$. A missing file is not measured.

D. Release score

The scalar in (3) is null for every method in this paper. The supplement lists that column and the invented arithmetic. The evaluation locks for a final test were not frozen.

E. BeyondArena

BeyondArena

(2ecfe882ccfb814fc27c4de10a64ceefd5d7655c) is not one of the 100 PMLB lineages and is not pooled with them.

5 of 12 families fit inside the cap. 7 did not prepare (4 outside the projected range, 3 overlapping). The range was not raised. Table XI has no retention column. The supplement names all 142 families.

VI. ABLATIONS

features12_steps8192 is the same residual with four times as many AdamW steps. It is a budget check, not the displayed model. Where both budgets have a complete three-seed group, the CatBoost median paired difference is 0.034 (68 lineages). The supplement gives the linear and MLP differences and lists all 100 lineages. A blank cell is an unavailable fit or an auditor that is not informative. It is not a zero. The scored

TABLE VII
GROUPED-VALIDATION FIDELITY AND EMPIRICAL PRIVACY. NOT DP OR HIPAA.

Method	α -prec. <i>Median [interval] (lineages).</i>	β -recall	DCR share (unbalanced)	DOMIAS
DOPE	0.905 [0.889, 0.918] (98)	0.437 [0.398, 0.473] (98)	0.811 [0.806, 0.814] (98)	0.508 [0.498, 0.520] (94)
Gaussian	0.812 [0.786, 0.832] (100)	0.430 [0.378, 0.453] (100)	0.806 [0.802, 0.811] (100)	0.513 [0.494, 0.527] (94)
Chow-Liu	0.858 [0.825, 0.904] (100)	0.359 [0.312, 0.400] (100)	0.810 [0.806, 0.816] (100)	0.512 [0.495, 0.522] (94)
Indep.	0.885 [0.868, 0.897] (100)	0.284 [0.197, 0.347] (100)	0.801 [0.797, 0.806] (100)	0.507 [0.496, 0.521] (94)
DOPE	0.902 [0.882, 0.940] (22)	0.441 [0.381, 0.489] (22)	0.811 [0.802, 0.819] (22)	0.508 [0.480, 0.538] (22)
CTGAN	0.719 [0.520, 0.856] (22)	0.170 [0.142, 0.267] (22)	0.820 [0.800, 0.842] (22)	0.502 [0.463, 0.534] (22)
TVAE	0.850 [0.731, 0.896] (22)	0.356 [0.283, 0.401] (22)	0.825 [0.813, 0.843] (22)	0.527 [0.487, 0.563] (22)

Contrast	α -prec.	β -recall	DCR share (unbalanced)	DOMIAS
<i>Line 2: Holm p; rank-biserial r; n. DOPE minus comparator.</i>				
	0.083 [0.045, 0.113]	0.024 [0.011, 0.039]	-0.0025 [-0.00625, 0.005]	0.00157 [-0.0069, 0.0074]
DOPE-Gaussian	1.86×10^{-10} ; 98 0.019 [0.00782, 0.056]	0.394; 98 0.048 [0.029, 0.072]	1.000; 0.000206; 98 -0.00125 [-0.00562, 0.00125]	1.000; -0.085; 94 0.000581 [-0.00802, 0.011]
DOPE-Chow-Liu	2.00×10^{-5} ; 98 0.018 [0.0072, 0.028]	1.32×10^{-10} ; 98 0.104 [0.068, 0.138]	0.806; -0.169; 98 0.00625 [0.0025, 0.011]	1.000; 0.00572; 94 0.00244 [-0.00595, 0.015]
DOPE-Indep.	5.47×10^{-5} ; 98 0.004; 98	9.71×10^{-15} ; 98 0.937; 98	0.005; 0.398; 98	1.000; 0.080; 94
DOPE-CTGAN	0.168 [0.054, 0.368] 0.004; 0.794; 22 0.071 [0.00713, 0.141]	0.219 [0.131, 0.256] 3.81×10^{-6} ; 1.000; 22 0.062 [0.00546, 0.135]	-0.00812 [-0.029, 0.015] 0.532; -0.277; 22 -0.025 [-0.030, -0.00641]	-0.00028 [-0.014, 0.031] 0.610; 0.130; 22 -0.021 [-0.036, -0.00972]
DOPE-TVAE	0.040; 0.644; 22	0.069; 0.573; 22	0.059; -0.613; 22	0.108; -0.510; 22

TABLE VIII
NEURAL BLOCK AT $4n$, NOT POOLED. W/T/L COUNTS LINEAGES.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	CTGAN	21	0.953 [0.896, 0.990]	-0.081 [-0.202, 0.103]	0.881 [0.702, 1.150]	21/0/0	5.72×10^{-6}
CatBoost	TVAE	21	0.953 [0.896, 0.990]	0.669 [0.539, 0.787]	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Linear	CTGAN	21	0.999 [0.996, 1.015]	-0.033 [-0.225, 0.080]	1.049 [0.916, 1.236]	21/0/0	5.72×10^{-6}
Linear	TVAE	21	0.999 [0.996, 1.015]	0.796 [0.605, 0.867]	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
MLP	CTGAN	19	1.026 [0.998, 1.263]	-0.149 [-0.471, 0.083]	1.079 [0.877, 2.030]	19/0/0	1.53×10^{-5}
MLP	TVAE	19	1.026 [0.998, 1.263]	0.753 [0.478, 0.852]	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}

TABLE IX
FOREST-FLOW CONFIRMATION AT $4n$, CATBOOST. NOT POOLED.

Lineage	DOPE bytes	DOPE	Forest-Flow bytes	Forest-Flow
1193_BNG_lowbwt	2,436	0.970	416,921,273	1.004
503_wind	3,846	0.996	637,056,689	1.016
582_fri_c1_500_25	5,670	0.909	1,085,155,062	0.883
596_fri_c2_250_5	1,698	0.807	234,776,554	0.968
feynman_l_12_1	1,032	0.991	123,784,681	1.001
strogatz_vdp1	984	0.581	115,202,253	1.036

TABLE X
PUBLISHED MATCHED RECEIPTS AT $4n$, CATBOOST.

Publication	Lineages	DOPE	Receipt	Difference
TabDDPM	not measured	not measured	not measured	not measured
TabSyn	not measured	not measured	not measured	not measured
Forest-Flow	not measured	not measured	not measured	not measured
ARF, author_default	97	0.940	0.840	0.077
ARF, native_selected	97	0.940	0.826	0.095

TABLE XI
BEYONDARENA FAMILIES THAT PREPARED. NO RETENTION AUDITOR.

Family	Fit rows	2,048 B	8,192 B	Val 2,048	Val 8,192
airfoil	802	804	818	0.010	0.00448
garments	939	2,317	2,372	0.028	0.025
musk	3,738	9,273	9,350	0.146	0.090
telemonitoring	257	1,852	1,890	0.033	0.019
video	35,918	5,054	5,071	0.00247	0.00195

fits did not keep a per-step series. The supplement plots the logged replay. It is not a substitute for the scored artifacts. The lower loss at 8,192 steps does not replace the displayed profile.

The primary informativeness threshold is 0.01 times the absolute null loss. Table XII lists the CatBoost median by threshold and the informative counts. The middle row is the 0.01 rule behind the retention tables. Thresholds 0, 0.001, and

0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Fit-seed variance across seeds other than 11 is not measured.

VII. DISCUSSION

CTGAN and TVAE do not match the CatBoost retention, and their successful size- $4n$ artifacts miss the cap. None of

TABLE XII
INFORMATIVENESS THRESHOLD AND AUDITOR COUNTS.

Threshold	DOPE	Gaussian	Chow–Liu	Indep.
0	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.001	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.01	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.05	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.1	0.942 (96)	0.668 (98)	0.584 (98)	-0.012 (98)

Threshold	CatBoost	Linear	MLP
0.001	97	93	75
0.01	97	92	75
0.05	97	90	73

these comparisons emits MFS-v2, PTF-v1, or a release at the 10,240-byte tier. A small Wilcoxon p -value does not flip that null. The 8,192-step budget moves the CatBoost median by 0.034 and is not the displayed model. BeyondArena fits 5 further families inside the cap and does not show retention. The unmeasured publications and the null release scores are unchanged from Section I-0a. An equal tuning budget across generators is not measured.

VIII. THREATS TO VALIDITY

a) Internal.: The fits use the coreset in Section IV, fit seed 11 only, and a training-derived validation cut of the official training rows. The scored retention panel is regression only. Other tasks are not a result of these tables. The supplement records the draw. The official test was not an input to training, selection, or the numbers in this paper. The informativeness rule drops a negligible real-versus-null gap. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Incomplete groups are omitted, not zeroed. Chow–Liu and independent marginals are implementations written for this comparison. CTGAN and TVAE use the lineages that finished. Runs that stopped at the recorded time limit are omitted. They are not counted as failures of a finished fit.

b) External.: These fits do not establish behavior on the full tables used by the diffusion papers, on a five-seed campaign, or on the official test. Generalization past seed 1729 is not measured. Forest-Flow’s byte comparison uses 6 lineages and the hash-checked fit files. BeyondArena is a separate fit check. It is not a retention panel, and it is not pooled with the density block.

c) Construct.: Retention is the reported utility. It is not the MFS-v2 scalar, which stays null. The linear auditor does not separate DOPE from the Gaussian copula, and that result is not an equivalence. Shipping this artifact is not the same experiment as shipping the compact predictor itself. The panel does not contain that control, so it does not show that a recipient needs the synthetic rows rather than the predictor. Normalized-space RMSE is the loss the auditors stored. Original-unit error after inverting the training transform was not stored, and this paper does not report it. Table VII is not a membership-inference certification. Its unbalanced DCR share and DOMIAS AUC are empirical. Empirical resemblance is not differential privacy and is not HIPAA de-identification.

d) Statistical.: The rank tests are descriptive, not the pre-registered production superiority test. Density, neural, Forest-Flow, and BeyondArena stay unpooled. A small retention p -value does not flip the null scalar.

IX. CONCLUSION

On 97 regression lineages and fit seed 11, generator `features12_steps2048` has CatBoost retention 0.940 (interval 0.914 to 0.976) and 98/100 artifacts inside the cap. CatBoost separates it from the density comparators. The linear auditor does not separate it from the Gaussian copula, and that result is not an equivalence. The 6-lineage Forest-Flow block leads on CatBoost (1/0/5) and the MLP (0/0/4). Those counts are not a certified win. The unmeasured publications and the release scores are unchanged from Section I-0a. Official tests stay sealed.

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